

INTERNSHIP PROJECT REPORT

ON

DocuTrust: Enterprise Advanced RAG Platform with Automated Self-Correction

Submitted in Partial Fulfillment of the Requirements for the Internship Project

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Introduction :

In today's digital world, organizations generate and store a large number of digital documents every day. These documents include company policies, employee records, legal agreements, financial reports, research papers, user manuals, and technical documentation. Although this information is valuable, finding the correct answer from hundreds of pages is often difficult and time-consuming. Employees usually spend a significant amount of time searching through multiple documents, which reduces productivity and delays decision-making.

Traditional document search systems mainly depend on keyword matching. They only search for exact words or phrases and are unable to understand the actual meaning of the user's question. If the required information is written using different words or appears in another section of the document, the search system may fail to provide accurate results. As a result, users often receive incomplete, irrelevant, or confusing information.

DocuTrust is an Artificial Intelligence (AI) powered document question-answering platform designed to solve this problem. It uses **Retrieval-Augmented Generation (RAG)** technology to understand user questions and search through uploaded documents intelligently. Instead of simply matching keywords, the system analyses the meaning of the question and retrieves the most relevant information before generating an answer. This approach significantly improves the accuracy and reliability of the responses.

One of the most important features of DocuTrust is its **Automated Self-Correction** mechanism. Before displaying an answer, the system verifies whether the retrieved information is relevant and complete. If the available information is insufficient, the system can improve the search process by rewriting the query and retrieving better results. This reduces incorrect answers and increases user trust in the system.

DocuTrust also provides **source citations** along with every answer. Users can easily identify which document and section were used to generate the response. This makes the platform more transparent, reliable, and suitable for professional environments where accuracy is very important.

The system is designed to be simple, secure, and user-friendly. Users only need to upload their documents and ask questions in natural language. The AI handles document analysis, information retrieval, answer generation, and source verification automatically. This saves time and improves productivity for both individuals and organizations.

DocuTrust can be used in many sectors such as education, healthcare, banking, law firms, government offices, and corporate companies. Any organization that manages a large collection of documents can benefit from this intelligent AI-powered assistant.

As Artificial Intelligence continues to improve, intelligent document management systems like DocuTrust will become an essential part of modern organizations. They help users access information quickly, reduce manual effort, improve decision-making, and ensure reliable access to important knowledge stored inside digital documents.

Problem Statement :

Many organizations store thousands of documents containing valuable information. These documents include policies, reports, contracts, research papers, employee manuals, and technical documents. Although all this information is available, finding the correct answer within a large collection of files is often difficult. Employees may spend several minutes or even hours searching manually, reducing productivity and increasing operational costs.

Traditional document search systems mainly rely on keyword matching. They cannot understand the actual meaning of a user's question and often return irrelevant results. If the exact keyword is missing from the document, the search system may fail even though the required information exists. This limitation creates confusion and makes information retrieval slow and inefficient.

Another major problem with many AI-based document assistants is **hallucination**. Sometimes AI generates answers that sound correct but are not actually present in the uploaded documents. This creates serious risks, especially in industries such as healthcare, banking, education, and legal services, where accurate information is essential.

Users also need confidence that the provided answer is based on genuine information. Without proper references or citations, it becomes difficult to verify the accuracy of AI-generated responses. This reduces user trust and limits the practical use of AI document assistants.

DocuTrust addresses these challenges by introducing an intelligent Retrieval-Augmented Generation (RAG) platform with Automated Self-Correction. The system searches uploaded documents semantically, verifies the retrieved information before generating answers, and provides proper citations for every response. This approach improves answer accuracy, reduces AI hallucinations, saves time, and creates a more reliable document search experience.

Objectives :

The primary objective of the DocuTrust project is to develop an intelligent AI-powered document management system that allows users to upload documents and receive accurate answers by asking questions in simple natural language. The system is designed to reduce the time and effort required to manually search through lengthy documents while improving the overall efficiency of information retrieval.

Another important objective is to improve the accuracy of document-based question answering. Instead of depending only on keyword matching, DocuTrust understands the meaning of the user's question and searches for relevant information using Artificial Intelligence. This helps users receive more meaningful and context-aware responses from the uploaded documents.

The project also aims to reduce AI hallucination by implementing a Retrieval-Augmented Generation (RAG) architecture with an automated self-correction mechanism. Before generating the final answer, the system verifies whether the retrieved information is relevant and sufficient. If necessary, it automatically improves the search process to provide a more reliable response.

One of the key objectives is to provide **transparent and trustworthy answers** by displaying proper document citations. Users can easily identify the exact document and section from which the answer has been generated. This increases confidence in the system and supports better decision-making in professional environments.

DocuTrust is also designed to improve productivity within organizations. Employees no longer need to manually read hundreds of pages to find specific information. Instead, they can simply upload documents, ask questions, and receive quick and accurate answers within seconds. This saves valuable time and allows employees to focus on more important tasks.

Another objective is to create a secure and user-friendly platform that can be used by people with different levels of technical knowledge. The interface is simple, allowing users to upload documents, type questions, and receive responses without any complex procedures. The project also ensures that uploaded documents remain organized and easily accessible.

Finally, the project aims to demonstrate the practical use of Artificial Intelligence in real-world document management. By combining AI, Machine Learning, Natural Language Processing, and modern web technologies, DocuTrust provides a smart solution that can be adopted by businesses, educational institutions, hospitals, banks, government organizations, and legal firms.

Methodology :

The DocuTrust project follows a structured and intelligent workflow to provide accurate document-based question answering. The process begins when a user uploads one or more documents such as PDF files, Word documents, reports, manuals, or policy files through the web application. The system securely stores these documents and prepares them for AI processing.

After the upload is completed, the system extracts the text from the uploaded documents. During this stage, unnecessary formatting, blank spaces, and unwanted symbols are removed. The extracted content is then divided into smaller meaningful sections called **document chunks**. Breaking large documents into smaller sections allows the AI model to retrieve information more efficiently and improves the accuracy of the search process.

Next, each document chunk is converted into numerical representations called **embeddings** using an embedding model. These embeddings capture the meaning of the text instead of simply storing individual words. The embeddings are then stored in a vector database, making it possible to perform semantic searches based on meaning rather than exact keyword matching.

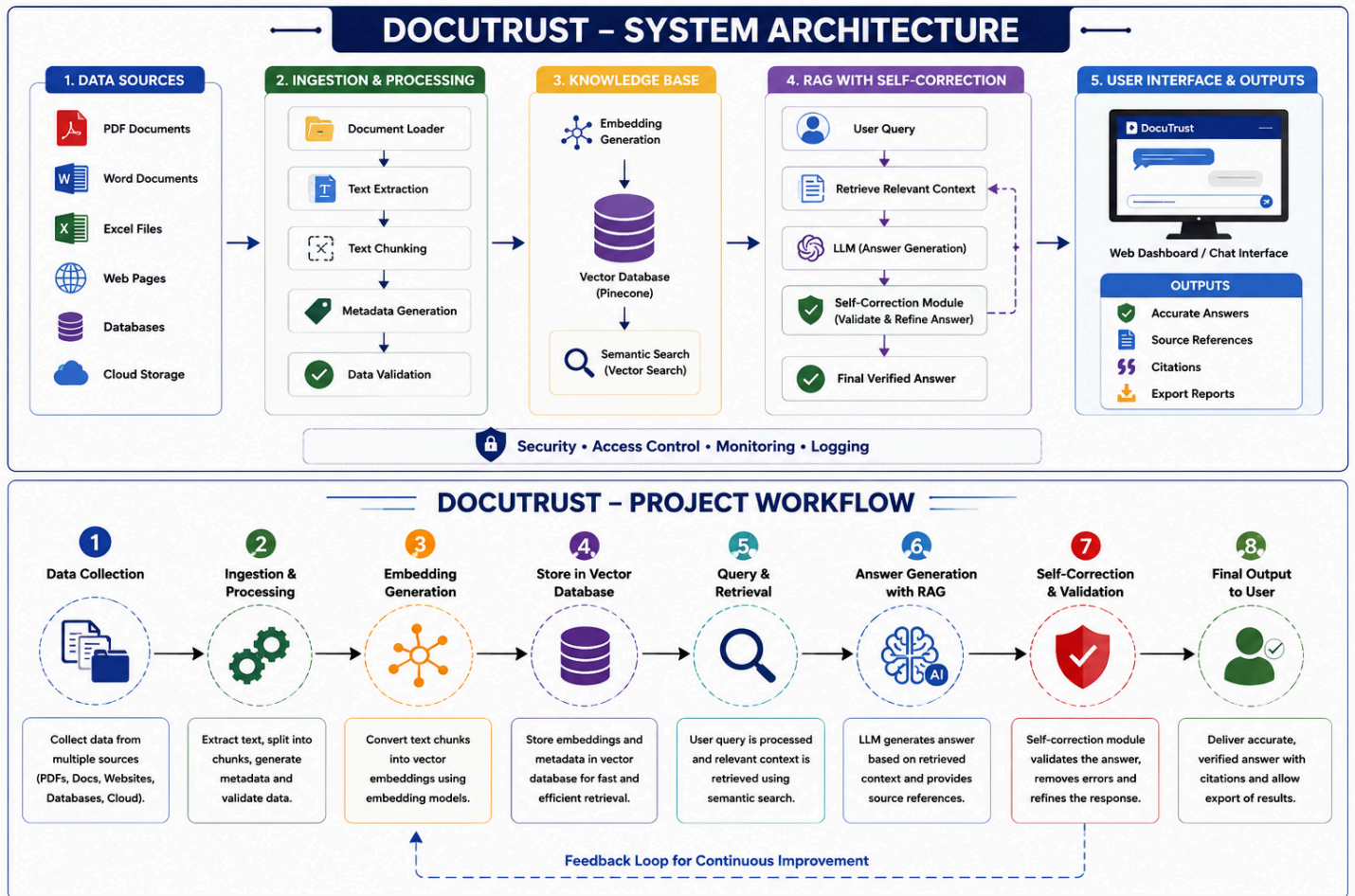
When the user asks a question, the same embedding process is applied to the user's query. The system compares the query embedding with the stored document embeddings and retrieves the most relevant document sections. Instead of searching only for matching keywords, the AI understands the context of the question and finds information with similar meaning.

After retrieving the relevant content, the **Automated Self-Correction** module checks whether the retrieved information is sufficient to answer the question accurately. If the retrieved content is incomplete or has a low confidence score, the system automatically rewrites the search query and performs another retrieval process. This additional verification step significantly reduces incorrect answers and improves overall reliability.

Once the best document sections have been identified, the Large Language Model (LLM) generates a final response using only the retrieved information. The answer is written in clear and understandable language while avoiding unsupported information. Along with the answer, the system displays the document name and citation so users can verify the source of the information.

The backend manages document uploads, embedding generation, vector storage, AI processing, and user requests, while the frontend provides a simple interface for document upload, question input, and answer display. The complete workflow ensures that DocuTrust delivers fast, accurate, secure, and trustworthy responses for users working with large collections of documents.

Suggested System Architecture and project workflow :



Technologies Used :

DocuTrust uses a combination of modern Artificial Intelligence, Machine Learning, Natural Language Processing, Web Development, and Database technologies to create a smart and reliable document question-answering system. Each technology plays an important role in ensuring that the platform provides accurate, secure, and efficient document retrieval.

1. **Python** : It is used as the primary programming language because it provides excellent support for Artificial Intelligence and Machine Learning development. It offers powerful libraries that simplify document processing, AI model integration, and backend development. Python also makes the system scalable and easy to maintain.
2. **FastAPI** : It is used as the backend framework. It manages document uploads, user authentication, AI processing, database communication, and API requests. FastAPI is known for its high performance and fast response time, making it suitable for applications that process multiple user requests simultaneously.
3. **LangGraph** or **CrewAI** : It can be used to manage multiple AI agents within the system. These frameworks help coordinate different tasks such as document retrieval, relevance checking, answer generation, and automated self-correction. Using multiple AI agents improves the quality and reliability of the final response.
4. **Large Language Models (LLMs)** : They are responsible for generating human-like answers based only on the retrieved document content. Instead of producing random responses, the language model uses the retrieved information to create clear, natural, and meaningful answers while maintaining accuracy.
5. **Embedding Models** : It convert both documents and user questions into numerical vectors that represent their meaning. These embeddings make semantic search possible, allowing the system to understand the context of a question rather than depending only on exact keyword matching.
6. **MongoDB** : It is used as the primary database to store uploaded documents, document metadata, user information, AI interaction history, and search records. Its flexible document structure makes it suitable for handling large collections of different document formats.
7. **HTML, CSS, and JavaScript** : They are used to develop the frontend interface. HTML provides the page structure, CSS creates an attractive and responsive design, and JavaScript enables interactive features such as document uploads, question submission, loading indicators, and real-time answer display.

The project can also use **Vector Databases** such as FAISS or ChromaDB for storing document embeddings. These databases allow fast semantic searching and significantly improve the efficiency of document retrieval, especially when handling thousands of documents.

Together, these technologies create a secure, intelligent, and user-friendly AI document assistant capable of providing fast and trustworthy answers from uploaded documents.

Advantages :

DocuTrust offers several advantages compared to traditional document search systems. One of its biggest benefits is that users can ask questions in simple natural language instead of searching manually through hundreds of pages. This makes information retrieval much faster and more convenient.

The project provides highly accurate answers by using Retrieval-Augmented Generation (RAG). Instead of generating responses based only on AI knowledge, the system retrieves relevant information directly from the uploaded documents before generating the answer. This significantly improves answer quality.

Another major advantage is the **Automated Self-Correction** feature. Before displaying the final response, the system checks whether the retrieved information is relevant and complete. If necessary, it automatically performs another search to improve the answer. This reduces AI hallucinations and increases reliability.

DocuTrust also improves transparency by providing document citations with every answer. Users can easily verify where the information came from, making the platform more trustworthy and suitable for professional environments.

The system saves a significant amount of time and increases productivity by reducing manual document searching. Employees, students, researchers, and professionals can quickly access important information without reading entire documents.

Another advantage is its scalability. The system can easily manage thousands of documents while maintaining fast search performance. It is suitable for both small organizations and large enterprises with extensive document collections.

Overall, DocuTrust provides accurate, reliable, secure, and intelligent document management while improving the overall user experience.

Applications :

DocuTrust has a wide range of practical applications across different industries. One of its most common uses is in **corporate organizations**, where employees frequently need to search company policies, employee handbooks, financial reports, technical manuals, and internal documents. The platform helps employees quickly find accurate information without manually reading lengthy files.

Educational institutions can also benefit from DocuTrust. Students can upload textbooks, lecture notes, research papers, and study materials to receive instant answers to academic questions. Teachers can use the system to organize educational content and simplify information retrieval for students.

Hospitals and healthcare organizations can use DocuTrust to manage medical guidelines, treatment protocols, research papers, and hospital policies. Doctors and healthcare professionals can quickly access important information while reducing the time spent searching through large medical documents.

Law firms and legal professionals can upload legal contracts, case files, government regulations, and legal documents into the platform. The AI assistant helps lawyers locate important clauses and legal information more efficiently, improving productivity and reducing manual effort.

Banks and financial institutions can use DocuTrust to organize compliance documents, financial regulations, audit reports, customer policies, and operational manuals. Employees can retrieve accurate information quickly while ensuring regulatory compliance.

Government departments can implement DocuTrust for managing public policies, official documents, administrative records, and legal notices. This helps government employees access verified information more efficiently while improving public service delivery.

Research organizations can use the platform to manage research papers, journals, technical documentation, and scientific publications. Researchers can easily search across multiple documents, saving valuable time during literature reviews and project development.

Overall, DocuTrust is a versatile AI-powered document assistant that can improve productivity, reduce manual work, and provide reliable information across various professional sectors.

Future Scope :

DocuTrust has significant potential for future development as Artificial Intelligence and document management technologies continue to evolve. One of the most valuable improvements would be the integration of **multilingual document support**. Users would be able to upload documents written in different languages and ask questions in their preferred language. The system would automatically understand both the document and the user's query, making it useful for international organizations and global businesses.

Another important enhancement is the implementation of **voice-based interaction**. Instead of typing questions, users could simply speak their queries using a microphone. The system would convert speech into text, process the request, and provide both spoken and written responses. This feature would improve accessibility for users who prefer voice communication or have physical limitations.

DocuTrust can also be expanded by integrating with **cloud storage platforms** such as Google Drive, Microsoft OneDrive, Dropbox, and SharePoint. Instead of uploading documents manually every time, users could directly access their cloud files, making document management more convenient and efficient.

Another future improvement is the development of a **real-time collaborative workspace**. Multiple users could work on the same document collection simultaneously while maintaining secure access permissions. Teams would be able to share knowledge, search documents together, and receive consistent AI-generated responses. This feature would be especially useful for corporate organizations, research teams, and educational institutions. The project can further include **advanced document summarization**. Instead of answering only specific questions, the system could automatically generate summaries of lengthy reports, research papers, legal contracts, and business documents. This would save users significant time when reviewing large amounts of information.

Future versions of DocuTrust may also include **document comparison features**. Users could upload two or more documents and ask the system to identify differences, similarities, policy changes, or missing information. This functionality would be extremely useful in legal firms, government departments, and corporate compliance teams.

Security can also be strengthened by introducing **role-based access control and encryption**. Different users could receive different permission levels based on their responsibilities. Sensitive documents could be protected using advanced encryption methods, ensuring that confidential information remains secure. Another exciting enhancement is the integration of **real-time web knowledge**. If users allow internet access, the system could combine information from uploaded documents with trusted online resources while clearly identifying the source of every piece of information. This would provide more comprehensive and updated responses.

Finally, DocuTrust has the potential to become a complete **Enterprise Knowledge Management System** capable of handling millions of documents, supporting multiple departments, providing intelligent search, generating automatic reports, and assisting organizations in making faster and more informed decisions. These future improvements will make DocuTrust an even more powerful AI-powered document assistant.

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7. FAISS Vector Search Documentation.
8. ChromaDB Documentation.
9. HTML, CSS, and JavaScript Official Documentation.
10. Python Official Documentation.

Conclusion :

DocuTrust is an intelligent AI-powered document question-answering platform designed to improve the way users search, understand, and manage digital documents. Unlike traditional keyword-based search systems, it uses Retrieval-Augmented Generation (RAG) technology to understand the meaning of user questions and retrieve the most relevant information from uploaded documents. This approach provides more accurate and context-aware responses while reducing the chances of incorrect or misleading answers.

One of the most important strengths of DocuTrust is its **Automated Self-Correction** mechanism. Before generating the final response, the system evaluates whether the retrieved information is sufficient and relevant. If necessary, it automatically performs additional searches to improve answer quality. This process increases user confidence and significantly reduces AI hallucinations.

The project also focuses on transparency by providing proper document citations with every answer. Users can easily verify the source of the information, making the system highly reliable for industries where accurate documentation is essential. This feature distinguishes DocuTrust from many traditional AI chat systems that generate answers without showing supporting evidence.

By combining Artificial Intelligence, Machine Learning, Natural Language Processing, and modern web technologies, DocuTrust offers a practical solution for businesses, educational institutions, healthcare organizations, legal firms, government departments, and research centres. It helps users save time, improve productivity, and make better decisions by providing fast and reliable access to important information.

The project demonstrates how modern AI technologies can transform document management into a smarter, more efficient, and user-friendly process. As organizations continue to generate increasing amounts of digital information, intelligent document assistants like DocuTrust will become essential tools for knowledge management and business operations.

In conclusion, DocuTrust is a modern, practical, and scalable AI solution that improves document search, enhances information accuracy, and increases organizational productivity. With future enhancements such as multilingual support, cloud integration, voice interaction, document summarization, and collaborative workspaces, the platform has excellent potential to become one of the most advanced enterprise document management systems.